



Full model of a buoyancy-driven airship and its control in the vertical plane

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ABSTRACT

A new concept of airship without thrust, elevator or rudder is considered in this paper. It is actuated by a moving mass and an internal air bladder with adjustable mass. This results into the motion of the center of gravity and the change of the net lift. The development of this concept of airship is motivated by energy saving. The moving mass and the airship's rigid body are viewed independently, which renders the modeling easy and displays clearly the interconnection of the two subsystems. An eight degrees-of-freedom (DOF) complete nonlinear mathematical model of this airship is derived through the Newton–Euler approach. The dynamics in the vertical plane are analyzed and controlled, using maximal feedback linearization. The modeling, analysis, and control are solved for special cases of the airship which become gradually complex. The most constrained case reduces to a two-DOF system. It is shown that the basic properties of this two-DOF mechanical system remain instrumental for the analysis and synthesis of advanced airships. These properties are far from being obvious when the most complex model is considered.

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1. Introduction

Although airships are no longer used for passenger transportation, they are useful for other purposes, such as advertising, sightseeing, monitoring and research [4,18,1]. Major countries in the world have independently set up new projects. For instance, Northrop Grumman's hybrid airship program [14], Lockheed Martin's program [3,12], Europe's ESA-HALE program [11,6], and some others [9,23].

The remarkable difference with ancient airships is that modern research concentrates on unmanned flying. Based on partial remote human control, feedback control and autonomous waypoint navigation, a modern airship is expected to complete some tasks automatically. Many works have been devoted to modeling, control, and trajectory tracking of airships [5,15,16].

Most of these works contributed to airships which are driven by propulsors along the airframe, and whose attitude is controlled by elevators and rudders or vector-propulsors. These mathematical models were derived from the Gomes model [5].

In this paper, a new-concept buoyancy-driven airship is considered, which moves forward by moving an internal point mass and varying the total mass of the airship. The motivation for this

research is to develop an energy saving flight with long airborne endurance.

To present, the only monograph available on the buoyancy-driven airship is the doctoral dissertation [16]. Therein, a demonstration which includes the change of the net buoyancy of the craft via a blower, the modification of the attitude, and generating a small forward motion is mentioned, and this mechanism had been proved to be valid. However, the technologies were not mature [16, p. 12]. This pioneering contribution studied the feasibility, the physical 2D model of this kind of airship, and derived a PID control for the system. The research was undertaken in an environment without any disturbance. It is concluded that this new kind of airship could fly efficiently when the angle of attack is small. This work also included the energy-efficiency comparison between this new-concept airship and the conventional one. Obviously, severe wind conditions may give the limits of the airship design and the minimal use of a standard propulsion may become mandatory.

The present paper investigates theoretical aspects only, and real experimentation is beyond its goals. Nevertheless, the main contributions are supported by a set of simulation results. More precisely, the contribution of this paper are:

1. The introduction of a complete 3D mathematical model for this new concept of airship.
2. The display of the fundamental structure hidden in the complex model of the airship.

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Nomenclature

α	the angle of attack	m_w	the center offset mass
θ	the pitch angle	$\bar{m}$	the mass of the moving mass
$\lambda_2, \lambda_1, \lambda_0$	parameters of the feedback controller	m_i	the i th element of $\mathbf{M}$
ξ	the flight path angle	m_0	the net buoyancy of the airship
ρ_a	the density of the air	m	the mass of the air displaced by the airship
∇	the volume of the airship	M_a	the aerodynamic pitch moment
$\boldsymbol{\Omega}$	the angular velocity of the airship in the body frame, $\boldsymbol{\Omega} \equiv (\Omega_1, \Omega_2, \Omega_3)^T$	$\mathbf{M}_a$	the aerodynamic moment, $\mathbf{M}_a \equiv \text{diag}\{L_a, M_a, N_a\}$
Ω_i	the i th element of $\boldsymbol{\Omega}$	$\mathbf{M}_{at}$	the aerodynamic moment with respect to the body frame
$\mathbf{b}$	the position of the airship in the inertial frame	$\mathbf{M}_{add}$	the inertia matrix
$\mathbf{B}$	the momentum of the added mass	$\mathbf{M}_f$	the added mass matrix, $\mathbf{M}_f \equiv \text{diag}\{m_{11}, m_{22}, m_{33}\}$
$\mathbf{B}_p$	the momentum of the moving mass, $\mathbf{B}_p \equiv (B_{p1}, B_{p2}, B_{p3})^T$	$\mathbf{M}_{GB}$	the resultant moments of gravity and buoyancy in the inertial frame
$\mathbf{B}_s$	the total momentum of the airship's rigid body	$\mathbf{M}_{GBt}$	the resultant moments of gravity and buoyancy in the body frame
C_i^j	aerodynamic coefficients	$\mathbf{M}_I$	the added inertial moment
$\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$	the body frame unit vectors	$\mathbf{M}_s$	the total external moment acting on the airship's rigid body
$\mathbf{F}_a$	the aerodynamic force, $\mathbf{F}_a \equiv \text{diag}\{X_a, Y_a, Z_a\}$	$\mathbf{M}_{int}$	the internal moment acting on the moving mass by the body
$\mathbf{F}_{at}$	the aerodynamic force with respect to the body frame	$\mathbf{M}$	$\mathbf{M} \equiv \text{diag}\{m_1, m_2, m_3\} = \mathbf{M}_f + m_s \mathbf{I}$ the total inertial mass matrix
$\mathbf{F}_I$	the added inertial force	$\mathbf{r}_b$	the position vector of m_b in the body frame
$\mathbf{F}_s$	the total external force acting on the airship's rigid body	$\mathbf{r}_p$	the position vector of the moving mass in the body frame, $\mathbf{r}_p \equiv (r_{p1}, r_{p2}, r_{p3})^T$
$\mathbf{F}_{int}$	the internal force acting on the moving mass by the body	$\mathbf{r}_w$	the position vector of m_w in the body frame
$\mathcal{I}$	the identity matrix of appropriate dimensions	$\mathbf{R}_1$	the rotation matrix from the body frame to the inertial frame
$\mathbf{F}_{GB}$	the resultant force of the gravity and the buoyancy in the inertial frame	$\mathbf{R}_2$	the rotation matrix from the velocity frame to the body frame
$\mathbf{F}_{GBt}$	the resultant force of the gravity and the buoyancy in the body frame	T_{add}	the kinetic energy of fluid disturbances
J	the second diagonal element of $\mathbf{J}_s$	u	the internal force applied on the moving mass along the $\mathbf{e}_1$ axis, $u \equiv u_1$
J_i	the i th diagonal element of $\mathbf{J}$	$\mathbf{u}$	the internal force acting on the moving mass by the airship, $\mathbf{u} \equiv (u_1, u_2, u_3)^T$
$\mathbf{J}_f$	the added inertial matrix, $\mathbf{J}_f \equiv \text{diag}\{m_{44}, m_{55}, m_{66}\}$	u_4	the control of the mass of the internal air bladder
$\mathbf{J}_s$	the moment of inertia of m_s , $\mathbf{J}_s \equiv \text{diag}\{J_x, J_y, J_z\}$	$\mathbf{v}$	the velocity vector of airship in the body frame, $\mathbf{v} \equiv (v_1, v_2, v_3)^T$
$\mathbf{J}$	the moment of inertia of m_s and the added mass, $\mathbf{J} = \text{diag}\{J_1, J_2, J_3\} = \mathbf{J}_f + \mathbf{J}_s$	$\mathbf{v}_p$	the velocity of the moving mass in the body frame
k	the parameter of the feedback controller	v_p	the velocity of the moving mass in the vertical plane
$\mathbf{i}, \mathbf{j}, \mathbf{k}$	the inertial frame unit vectors	X_a	the drag element in the aerodynamic force
$\mathbf{K}$	the angular momentum of the added mass	Z_a	the lift element in the aerodynamic force
$\mathbf{K}_s$	the total angular momentum of the airship's rigid body		
m_b	the mass of the internal air bladder		
m_h	the uniformly distributed mass of the hull		
m_s	the stationary mass of the airship		
m_v	the total mass of airship, $m_v = m_s + \bar{m}$		

3. The design of an advanced nonlinear control scheme based on this fundamental structure.

Based on the model of conventional airships, and inspired by the underwater glider, contribution 1 is obtained through Newton–Euler approach. In this process, the moving mass and the airship's rigid body are viewed as two independent parts. That makes it possible to refer to the modeling of the conventional airship [5], and also displays clearly the interconnection of the two subsystems. The complete 3D mathematical model of this kind of airship is complex and not available in the current literature. It is difficult to handle directly, even in the special case when the dynamics are restrained to the vertical plane.

Contribution 2 is derived starting from the simple case where the center of volume is fixed and ending with the general model subject to aerodynamic forces in the vertical plane. This way, essential properties are displayed and shown to be instrumental to derive an original nonlinear control scheme for the most complete

model. These fundamental properties of the airship are claimed to be invariant for the various special cases.

For this buoyancy-driven airship, an LQR approach and a simple nonlinear feedback had been used to control the dynamics, but the results had shown lots of limitations of these approaches [21, 22]. Thus, it is eager to find a suitable controller. Contribution 3 consists of the computation of a suitable output function for the most fundamental system such that it is minimum phase. Its feedback linearization achieves maximal linearization of the considered system and internal stability is ensured. This scheme is then used to design a controller for the complete model with similar performances.

Important assumptions are made in this paper. In these first results, only theoretical issues are argued and all states are assumed to be available. The disturbances, such as the wind, and variation of air density or of temperature, are not considered.

The paper is organized as follows. The buoyancy-driven airship is introduced in detail in Section 2. The airship is fully modeled in

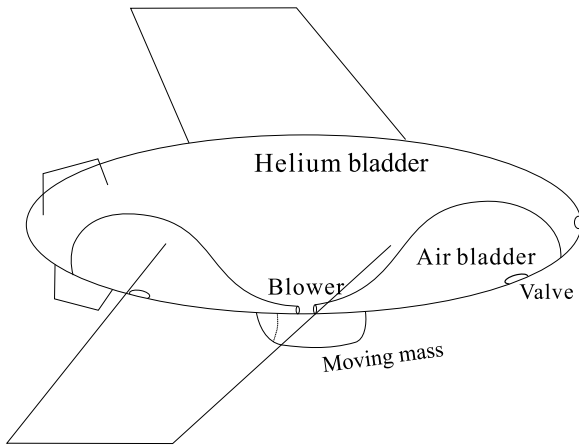


Fig. 1. Structure of buoyancy-driven airship.

Section 3. Section 4 contains the main system theoretic results derived for the airship system in the vertical plane. Four special cases are considered as the detailed analysis of the most constrained special case will be the key for the analysis of the general case. One major achievement of modern nonlinear control is feedback linearization [7,2]. Unfortunately the airship system is not fully linearizable, the main drawback is that a nonlinear dynamics will remain in the closed-loop system which may be unstable. In these cases a so-called non-minimum phase system has be dealt with. So, the internal stability is essential. In each case considered in Section 4, the maximal feedback linearization with internal stability is solved. Section 5 presents the conclusions of this paper.

The variables used in this paper are listed in the nomenclature.

2. The buoyancy-driven airship

The concept of this airship is derived from a combination of conventional airships [5] and underwater gliders [10], which replace traditional propulsion by a cyclic change of the net lift and the position of the center of gravity. This new concept of airship is first introduced and proven to be efficient in [16].

The basic structure is shown in Fig. 1.

2.1. The general description of this airship

The hull (or airframe) of the airship is considered to be ellipsoidal in this paper. The airship is inflated with helium and ambient air in two isolated bladders, shown in Fig. 1. The internal air bladder is symmetrical and elastic; its mass can be adjusted by a blower and valves. The blower is used to inflate the internal air bladder with ambient air, and the valves are used to release the air. The buoyancy of the airship is constant since the volume of the airship is invariant and the density of the atmosphere is assumed to be constant. By adjusting the mass of the air bladder, the net lift of the airship is changed, which leads the airship to rise or to fall.

There is a moving mass located at the bottom of the airship, and it can move on a bottom plane. According to the motion of the moving mass, the center of gravity of the airship CG also moves, which leads to the variation of the attitude of the airship.

The airship is also equipped with some fins which help to increase the aerodynamic forces, but there is neither rudder nor elevator.

2.2. The mathematical description of this airship

Two frames are attached to the airship: the inertial frame $G\{i, j, k\}$ and the body frame $O\{e_1, e_2, e_3\}$. The body frame

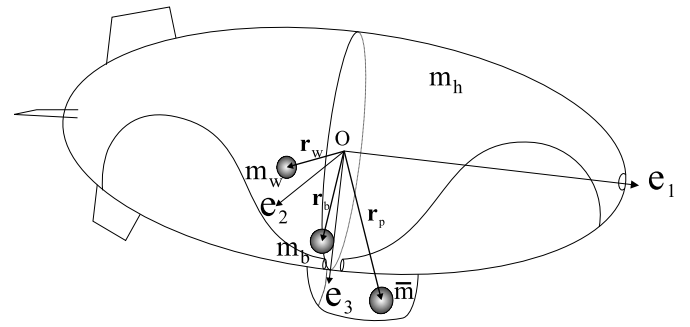


Fig. 2. Mass distribution.

$O\{e_1, e_2, e_3\}$ is attached to the reference point O at the geometrical center of the hull. The axis e_1 points to the prow, e_3 points to the bottom of the airframe, and e_2 is chosen to complete a right-handed Cartesian coordinate system. The inertial $G\{i, j, k\}$ frame is fixed to the earth, with the axis k along the direction of gravity.

The entire mass of the airship is split into several items. Let m_h denote the uniformly distributed mass of the hull, m_b denote the variable mass of the internal air bladder, $\bar{m}$ denote the mass of the moving mass, and m_w denote other internal fixed masses whose center of gravity offsets from O .

$\mathbf{r}_p \equiv (r_{p1}, r_{p2}, r_{p3})^T$ denotes the position of the moving mass in the body frame. $\mathbf{r}_b$ and $\mathbf{r}_w$ are the vectors from O to m_b and m_w , shown in Fig. 2. In the rest of this paper, it is assumed that m_b and m_w locate at point O , thus, $\mathbf{r}_b = \mathbf{r}_w = \mathbf{0}$.

Let m_s denote the total stationary mass, namely $m_s = m_h + m_b + m_w$. The total mass of the vehicle is represented by m_v , so:

$$m_v = m_s + \bar{m} = m_h + m_b + m_w + \bar{m}.$$

Let $m = \rho_a \nabla$ be the buoyancy of the airship with the volume ∇ , where ρ_a is the density of the ambient air. Let m_0 denote the net lift acting on the airship, thus, $m_0 = m_v - m$.

2.3. The operating mechanism of this airship

The mechanism to operate this kind of airship is described as follows. When releasing air from internal air bladder, the mass of the airship reduces, thus, the net buoyancy becomes positive and the airship rises. If the moving mass is driven by the airship's body to the rear at the same time, the airship gets a positive pitch angle θ , which yields a forward aerodynamic component force acting on the airship. This component force makes it move forward (see Fig. 3), and the trajectory is the BC segment of Fig. 5. Conversely, when pumping air into the internal air bladder, the mass of the airship increases. Thus, the net buoyancy becomes negative and the airship falls. With the moving mass driven by the body to the front, the pitch angle θ becomes negative, which still yields a forward aerodynamic component force. Therefore the airship moves downwards and forward (see Fig. 4), and the trajectory is the AB segment of Fig. 5. If the moving mass is moving to the sides, then the airship will roll. Due to the coupling of roll and rotation moments, the airship flies to the right or left directions [16]. Fig. 5 illustrates a typical trajectory of the airship in the vertical plane.

3. The mathematical modeling of the airship

The kinematics of the airship in the inertial frame is first presented. To derive the dynamic equations of the airship more easily and clearly, the airship is viewed as two independent subsystems: the body, and the moving mass. The airship's body is assumed as a rigid body, and the moving mass is viewed as a point mass, shown in Fig. 6. The coupling of these two subsystems only consists of

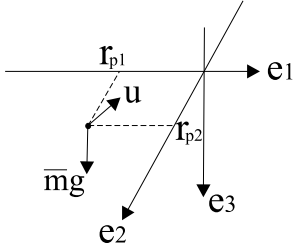


Fig. 7. The moving mass and the analysis of forces.

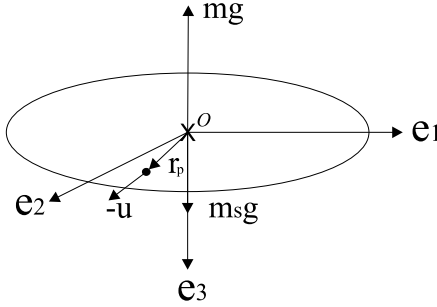


Fig. 8. The rigid body and some forces.

According to Newton's Second Law $\mathbf{F}_p = \bar{m} \dot{\mathbf{v}}_p$, with (2) and (3), the acceleration of the moving mass is derived as:

$$\ddot{\mathbf{r}}_p = -\dot{\mathbf{v}} - \dot{\boldsymbol{\Omega}} \times \mathbf{r}_p - \boldsymbol{\Omega} \times \dot{\mathbf{r}}_p + g(\mathbf{R}_1^T \mathbf{k}) + \mathbf{v}_p \times \boldsymbol{\Omega} + \mathbf{u}/\bar{m}. \quad (4)$$

Since r_{p3} is a constant, $\dot{r}_{p3} = 0$ and $\ddot{r}_{p3} = 0$. Thus, from (4), one gets:

$$u_3 = \bar{m}(\dot{v}_3 + \dot{\Omega}_1 r_{p2} - \dot{\Omega}_2 r_{p1} + \Omega_1(2\dot{r}_{p2} + v_2) - \Omega_2(2\dot{r}_{p1} + v_1) - g \cos \theta \cos \phi - \Omega_2^2 r_{p3} + \Omega_2 \Omega_3(r_{p2} + r_{p1}) - \Omega_1^2 r_{p3}). \quad (5)$$

3.3. The external forces and moments acting on the airship's rigid body

To derive the dynamic equations of the airship's rigid body through Newton–Euler approach, the force analysis is done on the rigid body.

Some forces acting on the rigid body are shown in Fig. 8. According to Newton's Third Law, for the rigid body, the impact from the moving mass is embodied by the force $-\mathbf{u}$.

Totally, the forces and moments acting on the body consist of inertial forces and moments, aerodynamic forces and moments, gravity and buoyancy, and internal forces and moments, which are explicitly presented in the following.

3.3.1. Inertial forces and added masses

The inertia of airships is much more significant due to its large Volume/Mass ratio in comparison with conventional airplanes. Therefore, inertial forces have to be considered.

In case of a rigid body in an ideal fluid with velocity v_i in the direction i , the force acting on the rigid body by the fluid in the direction j is $-m_{ij} \dot{v}_i$ and the parameter m_{ij} is called the added mass. There are 36 added masses for a rigid body in motion; the matrix $\mathbf{M}_{add}$ including these added masses is called the inertia matrix [17]:

$$\mathbf{M}_{add} = \begin{pmatrix} \mathbf{M}_f & * \\ * & \mathbf{J}_f \end{pmatrix}.$$

$\mathbf{M}_f$ is a symmetric 3×3 added mass matrix; $\mathbf{J}_f$ is an added inertia matrix; the other two blocks are added cross terms which

are neglected. Here, it is supposed that the airship is symmetric, so only elements on the diagonal are nonzero. Therefore, let $\mathbf{M}_f \equiv \text{diag}\{m_{11} \ m_{22} \ m_{33}\}$ and $\mathbf{J}_f \equiv \text{diag}\{m_{44} \ m_{55} \ m_{66}\}$.

In an ideal fluid, the kinetic energy T_{add} of fluid disturbances is

$$\begin{aligned} T_{add} &= \frac{1}{2} \sum_{i=1}^6 \sum_{j=1}^6 m_{ij} \epsilon_i \epsilon_j \\ &= \frac{1}{2} (m_{11} v_1^2 + m_{22} v_2^2 + m_{33} v_3^2 + m_{44} \Omega_1^2 \\ &\quad + m_{55} \Omega_2^2 + m_{66} \Omega_3^2) \end{aligned}$$

where $\epsilon_1 = v_1$, $\epsilon_2 = v_2$, $\epsilon_3 = v_3$, $\epsilon_4 = \Omega_1$, $\epsilon_5 = \Omega_2$, $\epsilon_6 = \Omega_3$. The momentum $\mathbf{B} \equiv (B_1, B_2, B_3)^T$, and the angular momentum $\mathbf{K} \equiv (B_4, B_5, B_6)^T$ of fluid disturbances are computed through the kinetic energy T_{add} as:

$$B_i = \frac{\partial T_{add}}{\partial \epsilon_i} \quad (i = 1, 2, \dots, 6).$$

So,

$$\mathbf{B} = \mathbf{M}_f \mathbf{v},$$

$$\mathbf{K} = \mathbf{J}_f \boldsymbol{\Omega}.$$

The inertial force $\mathbf{F}_I$ and moment $\mathbf{M}_I$ acting on the airship are as follows,

$$\begin{cases} \mathbf{F}_I = -\frac{d\mathbf{B}}{dt} = -\left(\frac{d\mathbf{B}}{dt} + \boldsymbol{\Omega} \times \mathbf{B}\right) = -\mathbf{M}_f \dot{\mathbf{v}} + (\mathbf{M}_f \mathbf{v}) \times \boldsymbol{\Omega}, \\ \mathbf{M}_I = -\frac{d\mathbf{K}}{dt} = -\left(\frac{d\mathbf{K}}{dt} + \boldsymbol{\Omega} \times \mathbf{K} + \mathbf{v} \times \mathbf{B}\right) \\ \quad = -\mathbf{J}_f \dot{\boldsymbol{\Omega}} + (\mathbf{J}_f \boldsymbol{\Omega}) \times \boldsymbol{\Omega} + (\mathbf{M}_f \mathbf{v}) \times \mathbf{v} \end{cases} \quad (6)$$

where $\frac{d\mathbf{B}}{dt}$, $\frac{d\mathbf{K}}{dt}$ denote the time-derivative of momentum $\mathbf{B}$ and angular momentum $\mathbf{K}$ with respect to the inertial frame, $\frac{d\mathbf{B}}{dt}$ and $\frac{d\mathbf{K}}{dt}$ denote the same variables in the body frame.

3.3.2. Aerodynamic forces and moments

Different pneumatic pressures are distributed on the surface of a vehicle flying in the atmosphere. The effect of these pneumatic pressures can be presented by aerodynamic forces $\mathbf{F}_a$ and moments $\mathbf{M}_a$ as follows,

$$\mathbf{F}_a \equiv (X_a, Y_a, Z_a)^T,$$

$$\mathbf{M}_a \equiv (L_a, M_a, N_a)^T.$$

X_a , Y_a , and Z_a are drag, sideforce, and lift, respectively. L_a , M_a , and N_a are roll moment, pitch moment, and yaw moment, respectively.

The aerodynamic forces act on the center of buoyancy O . By convention, the decomposed aerodynamic forces $\mathbf{F}_a$ lie in the velocity frame (also called the wind frame), and the moments $\mathbf{M}_a$ are decomposed in the body frame.

Since $\mathbf{F}_a$ is defined with respect to the velocity frame, $\mathbf{F}_a$ is rewritten with respect to the body frame and denoted by $\mathbf{F}_{at}$:

$$\mathbf{F}_{at} = \mathbf{R}_2 \mathbf{F}_a \quad (7)$$

where $\mathbf{R}_2$ is the transfer matrix from the velocity frame to the body frame. To keep the notations consistent, $\mathbf{M}_a$ is rewritten as $\mathbf{M}_{at}$.

For an airship with a volume ∇ , speed V , angle of attack α , and sideslip angle β , the forces, X_a , Y_a , and Z_a , and moments L_a , M_a , and N_a are displayed in approximate expressions as [19,15]:

$$\text{Drag: } X_a = \frac{1}{2} \rho_a \nabla^{2/3} V^2 (C_{x0} + C_x^\alpha \alpha^2);$$

$$\text{Sideforce: } Y_a = \frac{1}{2} \rho_a \nabla^{2/3} V^2 (C_{y0} + C_y^\beta \beta + C_y^{\Omega_3} \Omega_3);$$

$$\text{Lift: } Z_a = \frac{1}{2} \rho_a \nabla^{2/3} V^2 (C_{z0} + C_z^\alpha \alpha);$$

$$\text{Roll moment: } L_a = \frac{1}{2} \rho_a \nabla V^2 (C_{l0} + C_l^\beta \beta + C_l^{\Omega_1} \Omega_1 + C_l^{\Omega_3} \Omega_3);$$

$$\text{Pitch moment: } M_a = \frac{1}{2} \rho_a \nabla V^2 (C_{m0} + C_m^\alpha \alpha + C_m^{\Omega_2} \Omega_2);$$

$$\text{Yaw moment: } N_a = \frac{1}{2} \rho_a \nabla V^2 (C_{n0} + C_n^\beta \beta + C_n^{\Omega_1} \Omega_1 + C_n^{\Omega_3} \Omega_3).$$

Note from these expressions that the aerodynamic forces and moments are only dependent on the airship's velocities V and Ω_i , since the angle of attack α and the sideslip angle β are the functions of velocities. Thus, the aerodynamic forces and moments are controllable, but not directly actuated due to the absence of the control surface.

3.3.3. Gravity and buoyancy of the rigid body

For the airship's body (the moving mass excepted), its mass is m_s , mentioned in Section 2.2 and shown in Fig. 8. Besides the gravity, the buoyancy mg is acting on it. In the inertial frame, the composite effect of gravity and buoyancy is denoted by $\mathbf{F}_{GB}$ as follows,

$$\mathbf{F}_{GB} = (m_s - m)g\mathbf{k} = (m_0 - \bar{m})g\mathbf{k}$$

where $\mathbf{k}$ is a unit vector pointing in the direction of gravity. $\mathbf{F}_{GB}$ is transferred to the body frame as follows,

$$\mathbf{F}_{GBt} = (m_s - m)g\mathbf{R}_1^T \mathbf{k} = (m_0 - \bar{m})g\mathbf{R}_1^T \mathbf{k}. \quad (8)$$

It is assumed that the center of gravity of the rigid body locates at point O . So there is no moment involved by m_s and the buoyancy of the airship. In the body frame, the latter moment is:

$$\mathbf{M}_{GBt} = 0. \quad (9)$$

3.3.4. The internal forces $\mathbf{F}_{int}$ and moments $\mathbf{M}_{int}$

There exists a reacting force $-\mathbf{u}$ on the rigid body (see Fig. 8), and this is a significant difference from the conventional airship and airplane. These interactions $\mathbf{F}_{int}$ and $\mathbf{M}_{int}$ are given by:

$$\mathbf{F}_{int} = -\mathbf{u},$$

$$\mathbf{M}_{int} = \mathbf{r}_p \times (-\mathbf{u}).$$

3.4. The dynamic equations of the airship's rigid body

Let $\mathbf{B}_s$ and $\mathbf{K}_s$ be the translational and angular momenta of the airship's body, and they are expressed as,

$$\mathbf{B}_s = m_s \mathbf{v}, \quad (10)$$

$$\mathbf{K}_s = \mathbf{J}_s \boldsymbol{\Omega} \quad (11)$$

where $\mathbf{J}_s$ is the matrix of the moment of inertia of m_s of the rigid body.

Thus, the total force, denoted by $\mathbf{F}_s$, and the total moment, denoted by $\mathbf{M}_s$, acting on the body are derived through the above translational and angular momenta as follows:

$$\begin{aligned} \mathbf{F}_s &= \frac{d\tilde{\mathbf{B}}_s}{dt} = \frac{d\mathbf{B}_s}{dt} + \boldsymbol{\Omega} \times \mathbf{B}_s \\ &= m_s \dot{\mathbf{v}} + \boldsymbol{\Omega} \times (m_s \mathbf{v}), \end{aligned} \quad (12)$$

$$\begin{aligned} \mathbf{M}_s &= \frac{d\tilde{\mathbf{K}}_s}{dt} = \frac{d\mathbf{K}_s}{dt} + \boldsymbol{\Omega} \times \mathbf{K}_s + \mathbf{v} \times \mathbf{B}_s \\ &= \mathbf{J}_s \dot{\boldsymbol{\Omega}} + \boldsymbol{\Omega} \times (\mathbf{J}_s \boldsymbol{\Omega}) + \mathbf{v} \times (m_s \mathbf{v}). \end{aligned} \quad (13)$$

Here in (12), the time-derivative of the m_s is ignored since the magnitude of m_b which is the variable component of m_s is about 1/10 of the magnitude of m_s , and the variation of m_b in practice is slow.

From the force analysis in Section 3.3, $\mathbf{F}_s$ and $\mathbf{M}_s$ are the sum of the following terms:

$$\begin{aligned} \mathbf{F}_s &= \mathbf{F}_I + \mathbf{F}_{at} + \mathbf{F}_{GB} + \mathbf{F}_{int} \\ &= -\mathbf{M}_f \dot{\mathbf{v}} + (\mathbf{M}_f \mathbf{v}) \times \boldsymbol{\Omega} + \mathbf{F}_{at} + (m_0 - \bar{m})g\mathbf{R}_1^T \mathbf{k} - \mathbf{u}, \end{aligned} \quad (14)$$

$$\begin{aligned} \mathbf{M}_s &= \mathbf{M}_I + \mathbf{M}_{at} + \mathbf{M}_{GB} + \mathbf{M}_{int} \\ &= -\mathbf{J}_f \dot{\boldsymbol{\Omega}} + (\mathbf{J}_f \boldsymbol{\Omega}) \times \boldsymbol{\Omega} + (\mathbf{M}_f \mathbf{v}) \\ &\quad \times \mathbf{v} + \mathbf{M}_{at} + \mathbf{r}_p \times (-\mathbf{u}). \end{aligned} \quad (15)$$

The dynamic equations of the airship's body are derived from (12)–(15) as follows:

$$\dot{\mathbf{v}} = \mathbf{M}^{-1}(\mathbf{M} \mathbf{v} \times \boldsymbol{\Omega} + \mathbf{F}_{at} + (m_0 - \bar{m})g\mathbf{R}_1^T \mathbf{k} - \mathbf{u}), \quad (16)$$

$$\dot{\boldsymbol{\Omega}} = \mathbf{J}^{-1}(\mathbf{J} \boldsymbol{\Omega} \times \boldsymbol{\Omega} + \mathbf{M} \mathbf{v} \times \mathbf{v} + \mathbf{M}_{at} - \mathbf{r}_p \times \mathbf{u}) \quad (17)$$

where the mass matrix $\mathbf{M}$ and the moment-of-inertia matrix $\mathbf{J}$ are given by:

$$\begin{aligned} \mathbf{M} &= \mathbf{m}_s \mathcal{I} + \mathbf{M}_f = \text{diag}\{m_1 \quad m_2 \quad m_3\} \\ &= \text{diag}\{m_s + m_{11} \quad m_s + m_{22} \quad m_s + m_{33}\}, \end{aligned}$$

$$\begin{aligned} \mathbf{J} &= \mathbf{J}_s + \mathbf{J}_f = \text{diag}\{J_1 \quad J_2 \quad J_3\} \\ &= \text{diag}\{I_x + m_{44} \quad I_y + m_{55} \quad I_z + m_{66}\}. \end{aligned}$$

3.5. The 8-DOF mathematical model

As the airship is driven by change of the net lift, it is necessary to control the mass of the internal air bladder through the input u_4 , as

$$\dot{m}_b = u_4. \quad (18)$$

Combining with Eqs. (1), (4), (16), (17), and (18), the mathematical model of an 8-DOF buoyancy-driven airship is obtained as

$$\begin{pmatrix} \dot{\mathbf{b}} \\ \dot{\mathbf{R}}_1 \\ \dot{\mathbf{v}} \\ \dot{\boldsymbol{\Omega}} \\ \dot{\mathbf{r}}_p \\ \dot{\mathbf{T}} \\ \dot{m}_b \end{pmatrix} = \begin{pmatrix} \mathbf{R}_1 \mathbf{v} \\ \mathbf{R}_1 \hat{\boldsymbol{\Omega}} \\ \mathbf{M}^{-1} \bar{\mathbf{F}} \\ \mathbf{J}^{-1} \bar{\mathbf{K}} \\ \dot{\mathbf{r}}_p \\ \bar{\mathbf{T}} \\ u_4 \end{pmatrix} \quad (19)$$

where

$$\begin{aligned} \bar{\mathbf{K}} &= \mathbf{J} \boldsymbol{\Omega} \times \boldsymbol{\Omega} + \mathbf{M} \mathbf{v} \times \mathbf{v} + \mathbf{M}_{at} - \mathbf{r}_p \times \mathbf{u}, \\ \bar{\mathbf{F}} &= \mathbf{M} \mathbf{v} \times \boldsymbol{\Omega} + \mathbf{F}_{at} + (m_0 - \bar{m})g\mathbf{R}_1^T \mathbf{k} - \mathbf{u}, \\ \bar{\mathbf{T}} &= -\dot{\mathbf{v}} - \dot{\boldsymbol{\Omega}} \times \mathbf{r}_p - \boldsymbol{\Omega} \times \dot{\mathbf{r}}_p + g(\mathbf{R}_1^T \mathbf{k}) + \mathbf{v}_p \times \boldsymbol{\Omega} + \mathbf{u}/\bar{m}. \end{aligned}$$

4. The airship's dynamics in the vertical plane and its control

The dynamics of the airship in the vertical plane is considered since it is the most fundamental and important for the airship. It is not easy to design a nonlinear controller for the vertical planar dynamics due to complex nonlinearities. A common approach is to linearize the system around some operating points, and then design a linear controller for the linearized system. A weak point of this approach is the limitation of the feasibility of the linear controller for the real system due to the validity of the approximation.

To analyze the dynamics and to derive a nonlinear controller for the system, a progressive approach from a simplified model to a complete one is adopted. The simplest airship model corresponds to the special case in which the airship freely rotates around its fixed center of volume. The model is identical to the one of a prismatic-joint pendulum considered in the literature (see, e.g. [20]). Basic properties are derived and a control law which solves the maximal feedback linearization problem with internal stability is explicitly computed. By dropping some restrictions, two other less restrictive special models for the airship are considered. Although they are more complex, it is shown that the results obtained for the most simple special case remain valid, including for the complete model.

During the ascending and descending periods, the airship just adjusts the position of the moving mass to cope with general disturbances. Therefore, m_b is constant, and $u_4 = 0$ in the following analysis.

For several different special cases in this section, output functions which define a minimum phase system and whose feedback linearization will achieve maximal linearization are computed.

It is shown that the angular momentum of the airship plays a key role in these analysis and is instrumental for its full internal stabilization despite the intrinsic nonlinearity of the model.

Section 4.1 introduces some basic knowledge on feedback linearization and zero dynamics. Section 4.2 is devoted to the modeling and control of the most restrictive special case of the airship whose center of volume is fixed, and it is subject only to the motion of the moving mass. The airship is liberated in Section 4.3 by dropping the assumption that the center of volume O is fixed, which means the airship can ascend and descend. In this case, the center of mass is subject to a ballistic motion and the controllable subsystem is shown to keep similar properties as the previous fundamental system. In Section 4.4, the airship model is completed with the inertial forces and Section 4.5 displays the full model in the vertical plane including the aerodynamic forces. Up to some approximation, the previous system structure is still valid and used to design a nonlinear controller with similar performance. Section 4.6 presents simulation results of a continuing flight with the controller designed in Section 4.5.

4.1. Preliminaries on maximal feedback linearization and zero dynamics

A standard way to control a nonlinear system consists of looking for a nonlinear states feedback that fully, or partially, cancels out all nonlinear terms in some suitable coordinates. This has been one major achievement in the early years of nonlinear control theory. For completeness some main results that will be applied in the rest of this paper are now recalled.

Given an SISO nonlinear affine system,

$$\Sigma = \begin{cases} \dot{x} = f(x) + g(x)u, \\ y = h(x) \end{cases} \quad (20)$$

where the state $x \in \mathbb{R}^n$, the input $u \in \mathbb{R}$, the output $y \in \mathbb{R}$, and the entries of f , g are meromorphic functions. According to [2], if the relative degree of the output y is n , the system (20) can

be fully and exactly linearized by feedback. Pick new coordinates $\xi = (h, L_f h, \dots, L_f^{n-1} h)^T$, and system (20) is transformed into

$$\begin{cases} \dot{\xi}_1 = \xi_2, \\ \dot{\xi}_2 = \xi_3, \\ \vdots \\ \dot{\xi}_n = \alpha(\xi) + \beta(\xi)u, \\ y = \xi_1 \end{cases}$$

for some suitable $\alpha(\xi)$ and $\beta(\xi)$.

Define the nonlinear state feedback as:

$$u = \frac{1}{\beta(\xi)}(-\alpha(\xi) + v) \quad (21)$$

and the original nonlinear system is transformed into a linear system with transfer function $1/s^n$. By chosen $v = -\lambda_0 \xi_1 - \lambda_1 \xi_2 - \dots - \lambda_{n-1} \xi_n$, the characteristic polynomial of this linearized system is

$$s^n + \lambda_{n-1}s^{n-1} + \dots + \lambda_0 = 0$$

which is chosen to be stable. The overall nonlinear feedback for the system (20) is

$$u = \frac{1}{\beta(\xi)}(-\alpha(\xi) - \lambda_0 \xi_1 - \lambda_1 \xi_2 - \dots - \lambda_{n-1} \xi_n).$$

Whenever the relative degree of the system (20) is γ and $\gamma < n$, partial feedback linearization is obtained as follows. Pick

$$\xi_1 = h(x), \quad \xi_2 = L_f h(x), \quad \dots, \quad \xi_\gamma = L_f^{\gamma-1} h(x).$$

Since system (20) is single input, the left kernel $g^\perp$ of $g(x)$ has dimension $n - 1$, and it is fully integrable. Thus, $g^\perp$ can be written as

$$g^\perp = \text{span}\{d\xi_1, \dots, d\xi_{\gamma-1}; d\zeta\}$$

where $\dim \zeta = n - \gamma$. Consequently, in these new coordinates, system (20) reads

$$\begin{cases} \dot{\xi}_1 = \xi_2, \\ \vdots \\ \dot{\xi}_\gamma = \alpha(\xi, \zeta) + \beta(\xi, \zeta)u, \\ \dot{\zeta} = q(\xi, \zeta), \\ y = \xi_1. \end{cases} \quad (22)$$

The dynamics $\dot{\zeta} = q(\xi, \zeta)$ are known as the internal dynamics and whenever y is constrained to zero, then it yields the so-called zero dynamics [8]

$$\dot{\zeta} = q(0, \zeta).$$

The nonlinear feedback

$$u = \frac{1}{\beta(\xi, \zeta)}(-\alpha(\xi, \zeta) + v) \quad (23)$$

yields a linear transfer function from the new input v to the output y as $\frac{1}{s^\gamma}$. The feedback also yields the unobservable dynamics $\dot{\zeta} = q(\xi, \zeta)$. The output of system (20) is easily stabilized by pole placement as done in (23). However the internal stability is guaranteed if and only if the unobservable zero dynamics $\dot{\zeta}$ is stable. This is exactly what is called a minimum phase system and does depend on the choice of output only.

In the rest of the paper, an output will be sought such that its relative degree is maximal, and that it defines a minimum phase system.

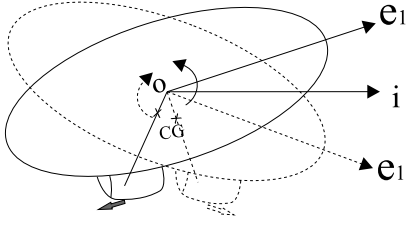


Fig. 9. The rotation of the airship with the movement of the moving mass.

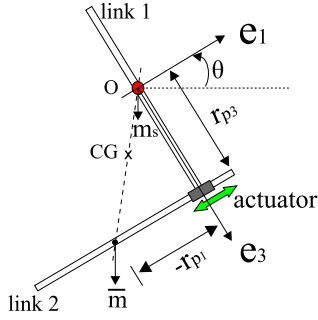


Fig. 10. The airship system with a fixed point.

4.2. The airship with fixed center of volume

It is clear that the dynamics in the vertical plane is impacted by the following four parts:

- (1) the moving mass which controls the pitch angle θ ,
- (2) the mass of the internal air bladder which controls the flying altitude,
- (3) the inertial forces,
- (4) the aerodynamic forces.

Assume that the airship is only subject to the motion of the moving mass to control the pitch angle, and there is no other force affecting the airship. In this situation, the moving mass actually impacts the angular momentum around the center point. The dynamics of the airship is not only restricted to the vertical plane, but it is also assumed that the center of volume O of the airship is fixed, which means that the airship can only rotate around O , and the longitudinal and horizontal translation velocities v_3 and v_1 are zero, as shown in Fig. 9.

These assumptions are now summarized:

Assumptions:

- A₁:** No aerodynamic force.
- A₂:** No inertial force.
- A₃:** The center of volume O is fixed, and the airship has no translational velocities.

The mathematical model of this fundamental case is derived as follows.

4.2.1. Model of the airship with a fixed center of volume

In the situation described above, the airship rotates around O depending on the forward and backward motions of the moving mass, as shown in Fig. 9, but O does not have any translation velocity.

In this case, the airship is identical to a prismatic-joint pendulum described in Fig. 10. The rotational joint at the point O is not actuated. The joint between the two links of the pendulum is prismatic and actuated. This pendulum, inverted or not, has been considered as a standard control example in many references [20].

Here, the rigid body of the airship is represented by the link 1 which is fixed at the point O , and the moving mass corresponds to the link 2 which can be moved by an actuator. The position of the center of the moving mass in the body frame in the vertical plane is (r_{p1}, r_{p3}) . The system can swing along with the actuator changing the position of $\bar{m}$, which is the same as the description in Section 2.3. Here, r_{p3} is constant, shown in Fig. 10.

For this case, $\mathbf{F}_s = 0$. The moving mass impacts the angular momentum, which is reflected by $\mathbf{M}_s = \mathbf{M}_{GBt}$ with the constraint that the velocity $\mathbf{v} = 0$. Combining Eqs. (4), (12), and (13), the model is derived as,

$$\begin{cases} \ddot{\theta} = \sigma_1 = -\frac{1}{J + \bar{m}r_{p1}^2} (\bar{m}r_{p3}r_{p1}\dot{\theta}^2 + 2\bar{m}r_{p1}\dot{r}_{p1}\dot{\theta} \\ \quad + \bar{m}gr_{p1}\cos\theta + r_{p3}u), \\ \ddot{r}_{p1} = \sigma_1 = \frac{1}{J + \bar{m}r_{p1}^2} \left((Jr_{p1} + \bar{m}r_{p1}^3 + \bar{m}r_{p3}^2r_{p1})\dot{\theta}^2 \right. \\ \quad + 2\bar{m}r_{p3}r_{p1}\dot{r}_{p1}\dot{\theta} - (J + \bar{m}r_{p1}^2)g\sin\theta \\ \quad \left. + \bar{m}gr_{p3}r_{p1}\cos\theta + \left(\frac{J}{\bar{m}} + r_{p1}^2 + r_{p3}^2 \right)u \right) \end{cases} \quad (24)$$

where the input u_1 and J_y are rewritten as u and J , respectively. ζ_1 and σ_1 are suitable notations which will be used later on.

4.2.2. Maximal linearization of a minimum phase model

System (24) is a special case of $\dot{x} = f(x) + g(x)u$ where $x = (\theta, \dot{\theta}, r_{p1}, \dot{r}_{p1})$. It is fully accessible as it can be checked from standard computations [7] that

$$\dim \text{span}\{g, \text{ad}_f g, \text{ad}_f^2 g\} = 4.$$

The distribution spanned by $\{g, \text{ad}_f g, \text{ad}_f^2 g\}$ is not involutive. Thus, system (24) is not fully linearizable by static state feedback. Then, it is interesting to look for a maximal linearization as done in [13]. Standard computations allow to check that the distribution spanned by $\{g, \text{ad}_f g\}$ is involutive. As done in [13], any state variable $h(x)$ such that $dh(x) \perp \{g, \text{ad}_f g\}$ will define an output whose relative degree is 3, and whose feedback linearization yields a linear third-order subsystem, and first-order zero dynamics.

According to the notations in [2] the following results are obtained from system (24):

$$\mathcal{H}_\infty = 0,$$

$$\mathcal{H}_3 = \text{span}\{d\phi_1, d\phi_2\}$$

and $\mathcal{H}_4$ is not integrable. The condition $\mathcal{H}_\infty = 0$ means that the system is fully accessible. $\mathcal{H}_3$ represents the codistribution which consists of all differential forms whose relative degree is at least 3. Here, ϕ_1 and ϕ_2 are computed as,

$$\begin{aligned} \phi_1 &= J\dot{\theta} + (r_{p1}^2\dot{\theta} + r_{p3}^2\dot{\theta} + r_{p3}\dot{r}_{p1})\bar{m}, \\ \phi_2 &= \theta + \frac{r_{p3}}{\sqrt{\frac{J}{\bar{m}} + r_{p3}^2}} \arctan \frac{r_{p1}}{\sqrt{\frac{J}{\bar{m}} + r_{p3}^2}}. \end{aligned}$$

The mechanical interpretation of those functions is as follows. In ϕ_1 , $J\dot{\theta}$ is the angular momentum of the static mass of the airship in the vertical plane, and $(r_{p1}^2\dot{\theta} + r_{p3}^2\dot{\theta} + r_{p3}\dot{r}_{p1})\bar{m}$ is a vertical plane component of the angular momentum of the moving mass $\mathbf{r}_p \times \bar{m}\mathbf{v}_p$. So, ϕ_1 is the angular momentum of the whole airship computed at the point O . Note that ϕ_1 is different from $\mathbf{K}$ in Section 3.3.1 which is the angular momentum of the inertial forces. ϕ_1 and ϕ_2 also have the following relation:

$$\dot{\phi}_2 = \frac{1}{J + (r_{p1}^2 + r_{p3}^2)\bar{m}}\phi_1. \quad (25)$$

Up to some integrating factor, ϕ_2 represents an integral of the angular momentum ϕ_1 .

At this stage, any combination of ϕ_1 and ϕ_2 has relative degree 3 and its feedback linearization will yield a linear controllable three-dimensional subsystem with a one-dimensional zero dynamics. The following result shows the possibility to ensure that the system is minimum phase which has a decisive impact on its internal stability and the feasibility of this control design.

Theorem 1. *The system (24) with the output $y = \phi_1 + k\phi_2$ has asymptotically stable zero dynamics for any $k > 0$.*

Proof. Under this situation, a new coordinate system is chosen as follows

$$\xi_1 = \phi_1 + k\phi_2, \quad (26)$$

$$\xi_2 = \dot{\phi}_1 + k\dot{\phi}_2, \quad (27)$$

$$\xi_3 = \ddot{\phi}_1 + k\ddot{\phi}_2, \quad (28)$$

$$\zeta = \phi_1. \quad (29)$$

From a practical point of view, the right-hand side of Eqs. (26)–(29) are computed using a computer algebra software, as the lengthy expression of those functions is trackable by computers. Their result is directly plugged in the simulation system for practical tests. For instance, Eqs. (26) to (28) read:

$$\begin{cases} \xi_1 = \phi_1 + k\phi_2 = J\dot{\theta} + (r_{p1}^2\dot{\theta} + r_{p3}^2\dot{\theta} + r_{p3}\dot{r}_{p1})\bar{m} \\ \quad + k\left(\theta + \frac{r_{p3}}{\sqrt{\frac{J}{\bar{m}} + r_{p3}^2}} \arctan \frac{r_{p1}}{\sqrt{\frac{J}{\bar{m}} + r_{p3}^2}}\right), \\ \xi_2 = -mg(r_{p1}\cos\theta + r_{p3}\sin\theta) \\ \quad + \frac{k}{J + mr_{p1}^2 + mr_{p3}^2}(J\dot{\theta} + mr_{p1}^2\dot{\theta} + mr_{p3}^2\dot{\theta} + mr_{p3}\dot{r}_{p1}), \\ \xi_3 = -mgr_{p1}\cos\theta + mgr_{p1}\dot{\theta}\sin\theta - mgr_{p3}\dot{\theta}\cos\theta \\ \quad - k(mr_{p1}^3g\cos\theta + 2mr_{p1}^2\dot{r}_{p1}\dot{\theta} + mr_{p3}g\sin\theta r_{p1}^2 \\ \quad + mr_{p1}g\cos\theta r_{p3}^2 + 2mr_{p1}\dot{r}_{p1}r_{p3} + 2mr_{p1}\dot{r}_{p1}\dot{\theta}r_{p3}^2 \\ \quad + mgr_{p3}^3\sin\theta + r_{p1}gJ\cos\theta + 2r_{p1}\dot{r}_{p1}\dot{\theta}J \\ \quad + gr_{p3}\sin\theta J)m/(J^2 + 2Jmr_{p1}^2 + 2Jmr_{p3}^2 \\ \quad + m^2r_{p1}^4 + 2m^2r_{p1}^2r_{p3}^2 + m^2r_{p3}^4). \end{cases} \quad (30)$$

The zero dynamics under these new coordinates is given by $\dot{\zeta}$ under the constraint $\phi_1 + k\phi_2 = 0$. Thus, $\dot{\phi}_1 = -k\dot{\phi}_2$. From (25) one gets

$$\dot{\zeta} = \dot{\phi}_1 = -k\dot{\phi}_2 = -k \frac{1}{J + (r_{p1}^2 + r_{p3}^2)\bar{m}}\phi_1.$$

Thus,

$$\dot{\zeta} = -k \frac{1}{J + (r_{p1}^2 + r_{p3}^2)\bar{m}}\zeta.$$

So, for any $k > 0$, the system is asymptotically stable as $J + (r_{p1}^2 + r_{p3}^2)\bar{m}$ is strictly positive and bounded. $\square$

Table 1
Physical properties of the airship.

Terms	Values	Terms	Values	Terms	Values
$\bar{m}$	30 kg	J	8000 kg m ²	C_{x0}	0.059
m_s	269 kg	ρ_a	1.29 kg/m ³	C_x^α	0.016
m_1	400 kg	∇	296 m ³	C_{x0}	0
m_3	500 kg	C_m^α	0.255	C_z^α	1.269
r_{p3}	2 m	$C_m^{\Omega_2}$	0	C_{m0}	0

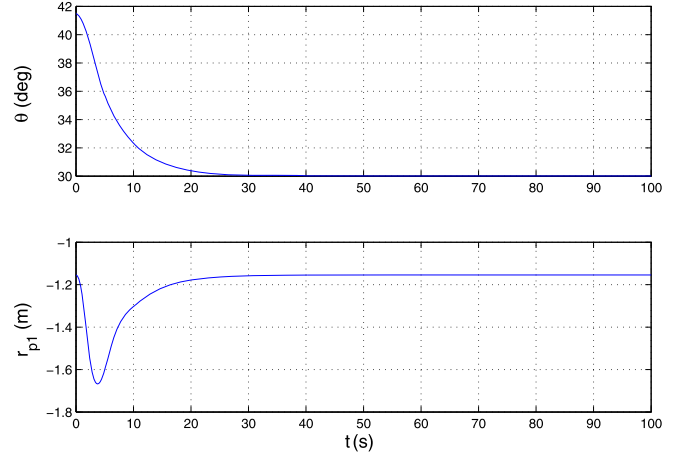


Fig. 11. Dynamics of θ and r_{p1} with initial error.

4.2.3. Control design and simulation

The family of outputs $y = \phi_1 + k\phi_2$, for a varying real number k , is now considered. The special case (24) is stabilized through standard input–output linearization according to Section 4.2.2. From Theorem 1, it is mandatory to pick $k > 0$ to ensure internal stability of the closed-loop system. Its actual value is a tuning parameter which influences the velocity of the zero dynamics. According to the feedback design in Section 4.1, the following error equation is considered,

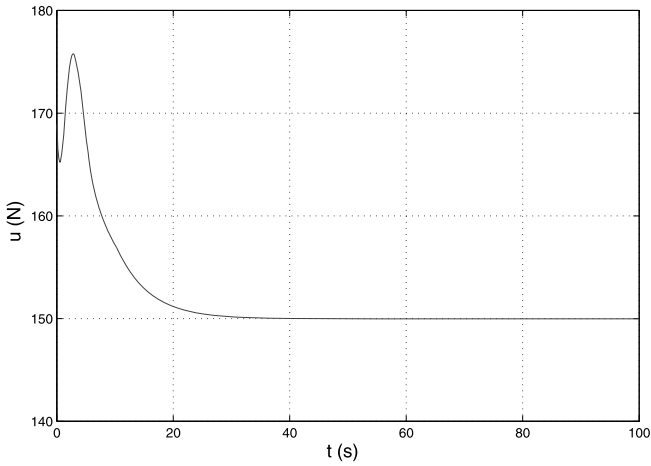
$$y^{(3)} + \lambda_2 y^{(2)} + \lambda_1 y^{(1)} + \lambda_0 (y - y_e) = 0 \quad (31)$$

where y_e is the desired constant value for y ; $y = \xi_1$; $y^{(1)} = \xi_2$; and $y^{(2)} = \xi_3$. Here, ξ_i are computed by Eq. (30). The term $y^{(3)}$ is an explicit function of the control u as $\dot{\xi}_\gamma$ in Section 4.1, which can be rewritten as $y^{(3)} = \alpha(\xi, \zeta) + \beta(\xi, \zeta)u$. Substituting the model (24) into the first-order time-derivative of $y^{(2)}$, these $\alpha(\xi, \zeta)$ and $\beta(\xi, \zeta)$ are derived by some computer algebra software. The computation may be time-consuming, but there is no drawback since they are performed offline. Actually, they deliver long mathematical expressions and are useless for practitioners as the feedback law is directly plugged into the control device. The control u is detailed in Appendix A of the paper and reads as:

$$u = -\frac{1}{\beta(\xi, \zeta)}(\lambda_2 \xi_3^{(2)} + \lambda_1 \xi_2^{(1)} + \lambda_0 (\xi_1 - y_e) + \alpha(\xi, \zeta)). \quad (32)$$

For these simulations, parameters $k = 50$, $\lambda_2 = 2$, $\lambda_1 = 2$, $\lambda_0 = 1$. The initial values of θ , $\dot{\theta}$, r_{p1} , and $\dot{r}_{p1}$ are 41.5°, 0, −1.15 m, and 0, respectively, and their commanded values are 30°, 0, −1.15 m, and 0, respectively. All mechanical properties used in the simulations are listed in Table 1 where the aerodynamic coefficients cite from [15].

Figs. 11 and 12 are simulation results of the dynamics of θ , r_{p1} , and the control input u . The settling times of θ and r_{p1} are around 20 s, which means that the airship quickly recovers to stable. There is no overshoot for the pitch angle under the control, and that of the r_{p1} is acceptable.

Fig. 12. The behavior of input u .

4.3. The airship with liberated center point

Aircrafts, as helicopters, rotate around the center of mass, not around the center of volume. In the first fundamental special case, the aircraft is assumed to rotate around the center of the volume O . So, by liberating the pivot O , the airship will rotate around the center of the gravity CG. The body frame is still attached to the point O .

For this case, assumption **A**₃ is dropped.

4.3.1. Model of the airship with liberated center point

In this case, only the gravity and the buoyancy are applied on the airship, which means that the right sides of Eqs. (14) and (15) only include the gravity and the buoyancy terms. Eqs. (12) and (13) remain unchanged,

$$\mathbf{F}_s = \mathbf{F}_{GBt}, \quad (33)$$

$$\mathbf{M}_s = \mathbf{M}_{GBt}. \quad (34)$$

Combining (4), (12), (13), (33), and (34), the mathematical model in the vertical plane is as follows,

$$\begin{cases} \ddot{\theta} = \varrho_1 + \frac{1}{J + \bar{m}r_{p1}^2}(\dot{v}_3 - \dot{\theta}v_1)\bar{m}r_{p1}, \\ \ddot{r}_{p1} = \sigma_1 + \frac{1}{J + \bar{m}r_{p1}^2}(\bar{m}r_{p1}r_{p3}(\dot{\theta}v_1 - \dot{v}_3) \\ \quad - (J + \bar{m}r_{p1}^2)(\dot{\theta}v_3 + \dot{v}_1)), \\ \dot{v}_1 = \frac{1}{m_s}(-m_s\dot{\theta}v_3 + (\bar{m} - m_0)g \sin \theta - u), \\ \dot{v}_3 = \frac{1}{m_s + \bar{m}}((m_s + \bar{m})\dot{\theta}v_1 + m_0g \cos \theta + \bar{m}r_{p3}\dot{\theta}^2 \\ \quad + 2\bar{m}\dot{\theta}r_{p1} + \bar{m}r_{p1}\ddot{\theta}). \end{cases} \quad (35)$$

Comparing the special case (24) with model (35), it is easy to note that system (24) is a subsystem of the model (35). The additional parts in $\ddot{\theta}$ and $\ddot{r}_{p1}$ are due to the translational velocities of point O .

4.3.2. Maximal linearization of a minimum phase model

System (35) is now subject to a ballistic motion (the point CG). In other words, there exist some non-controllable states besides the controllable or accessible subsystem, which is explained in the following. Again, following standard computation [2], one computes the non-controllable subsystem whose coordinates are denoted by ψ_1 and ψ_2 :

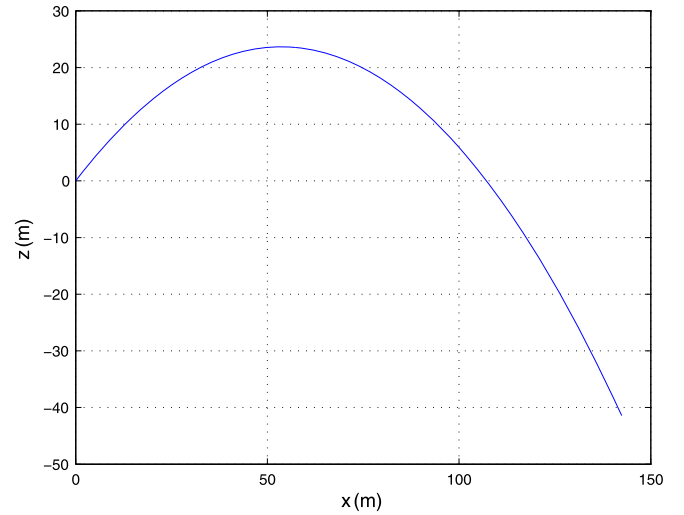


Fig. 13. Ballistic motion of CG.

$$\mathcal{H}_\infty = \text{span}\{d\psi_1, d\psi_2\}$$

where

$$\begin{aligned} \psi_1 &= \left(r_{p1}\dot{\theta} - \frac{m_s + \bar{m}}{\bar{m}}v_3\right)^2 + \left(\dot{r}_{p1} + r_{p3}\dot{\theta} + \frac{m_s + \bar{m}}{\bar{m}}v_1\right)^2, \\ \psi_2 &= \left(r_{p1}\dot{\theta} - \frac{m_s + \bar{m}}{\bar{m}}v_3\right)\cos \theta \\ &\quad + \left(\dot{r}_{p1} + r_{p3}\dot{\theta} + \frac{m_s + \bar{m}}{\bar{m}}v_1\right)\sin \theta. \end{aligned}$$

$(m_s + \bar{m})\psi_1$ is the kinetic energy of CG, and ψ_2 denotes the horizontal velocity of CG, which means that the CG of the airship is subject to a ballistic motion, shown in Fig. 13.

Thus the six-dimensional state system (35) can be decoupled into a two-dimensional non-controllable subsystem and a four-dimensional subsystem whose structural properties are similar to those of model (24). One gets

$$\mathcal{H}_3 = \text{span}\{d\tilde{\phi}_1, d\tilde{\phi}_2\} \oplus \mathcal{H}_\infty$$

where

$$\begin{aligned} \tilde{\phi}_1 &= J\dot{\theta} + (r_{p1}^2\dot{\theta} + r_{p3}^2\dot{\theta} + r_{p3}\dot{r}_{p1})\frac{\bar{m}m_s}{\bar{m} + m_s}, \\ \tilde{\phi}_2 &= \theta + \frac{r_{p3}}{\sqrt{\frac{\bar{m}+m_s}{\bar{m}m_s}J + r_{p3}^2}} \arctan \frac{r_{p1}}{\sqrt{\frac{\bar{m}+m_s}{\bar{m}m_s}J + r_{p3}^2}}. \end{aligned}$$

Here, $\tilde{\phi}_1$ is exactly the angular momentum of the airship computed at the center of gravity CG. The relation of $\tilde{\phi}_1$ and $\tilde{\phi}_2$ is as follows,

$$\dot{\tilde{\phi}}_2 = \frac{1}{J + (r_{p1}^2 + r_{p3}^2)\frac{\bar{m}m_s}{\bar{m} + m_s}}\tilde{\phi}_1.$$

Theorem 2. The system (35) with the output $\tilde{\phi}_1 + \tilde{k}\tilde{\phi}_2$ has asymptotically stable zero dynamics for any $\tilde{k} > 0$.

The proof is similar to that of Theorem 1.

4.3.3. Control design and simulation

The principle to derive the control is the same as in Section 4.2.3. Here, define the output $y = \tilde{\phi}_1 + \tilde{k}\tilde{\phi}_2$, and substitute $y^{(1)} = \dot{\tilde{\phi}}_1 + \tilde{k}\dot{\tilde{\phi}}_2$, $y^{(2)} = \ddot{\tilde{\phi}}_1 + \tilde{k}\ddot{\tilde{\phi}}_2$, $y^{(3)} = \dddot{\tilde{\phi}}_1 + \tilde{k}\dddot{\tilde{\phi}}_2$ into Eq. (31), the control u is solving from that equation through computer algebra.

Due to the same reason as previous case, these explicit expressions are not displayed.

For the simulations, the initial values of v_1 , $\dot{v}_1$, v_3 , and $\dot{v}_3$ are 1.8 m/s, 0, 0, and 0. Let $m_0 = 1$ kg. Other parameters remain the same values. As the previous simulation result, similar dynamic behaviors of θ , r_{p1} and control input u are obtained and are not reproduced here. The ballistic motion of CG is shown in Fig. 13.

4.4. The airship with liberated center and added masses

The model in Section 4.3 is completed now by including the inertial forces. Assumptions **A₂** and **A₃** are dropped for this case.

In this case, the angular momentum will be affected by the inertial forces. The results from previous sections are extended and adapted to this more complete case.

4.4.1. Model of airship with liberated center and added masses

In this situation, $\mathbf{F}_I$ and $\mathbf{M}_I$ are included in $\mathbf{F}_s$ and $\mathbf{M}_s$, which means that Eqs. (14) and (15) are modified as follows,

$$\mathbf{F}_s = \mathbf{F}_I + \mathbf{F}_{GBt}, \quad (36)$$

$$\mathbf{M}_s = \mathbf{M}_I + \mathbf{M}_{GBt}. \quad (37)$$

Combining (4), (12), (13), (36), and (37), the model of the airship subject to the inertial forces is as follows,

$$\begin{cases} \ddot{\theta} = \varrho_2, \\ \ddot{r}_{p1} = \sigma_2, \\ \dot{v}_1 = \kappa_1, \\ \dot{v}_3 = \kappa_3 \end{cases} \quad (38)$$

where

$$\begin{aligned} \varrho_2 &= -\frac{1}{J_2 + \bar{m}r_{p1}^2} (\bar{m}r_{p3}r_{p1}\dot{\theta}^2 + 2\bar{m}r_{p1}\dot{r}_{p1}\dot{\theta} + \bar{m}gr_{p1}\cos\theta \\ &\quad + r_{p3}u - (\dot{v}_3 - \dot{\theta}v_1)\bar{m}r_{p1}), \\ \sigma_2 &= \frac{1}{J_2 + \bar{m}r_{p1}^2} \left((J_2r_{p1} + \bar{m}r_{p1}^3 + \bar{m}r_{p3}^2r_{p1})\dot{\theta}^2 + 2\bar{m}r_{p3}r_{p1}\dot{r}_{p1}\dot{\theta} \right. \\ &\quad - (J_2 + \bar{m}r_{p1}^2)g\sin\theta + \bar{m}gr_{p3}r_{p1}\cos\theta \\ &\quad + \left(\frac{J_2}{\bar{m}} + r_{p1}^2 + r_{p3}^2 \right) u + \bar{m}r_{p1}r_{p3}(\dot{\theta}v_1 - \dot{v}_3) \\ &\quad \left. - (J_2 + \bar{m}r_{p1}^2)(\dot{\theta}v_3 + \dot{v}_1) \right), \\ \kappa_1 &= \frac{1}{m_1} (-m_3\dot{\theta}v_3 + (\bar{m} - m_0)g\sin\theta - u), \\ \kappa_3 &= \frac{1}{m_3 + \bar{m}} ((m_1 + \bar{m})\dot{\theta}v_1 + m_0g\cos\theta + \bar{m}r_{p3}\dot{\theta}^2 \\ &\quad + 2\bar{m}\dot{\theta}r_{p1} + \bar{m}r_{p1}\ddot{\theta}). \end{aligned}$$

Note that (35) and (38) have the same form, with only J and m_s in (35) modified to J_2 , m_1 and m_3 in (38), and the other items remain the same. Therefore, model (38) reduces to the model (35) if $m_{11} = m_{33} = J_{22} = 0$ (no added mass).

4.4.2. Maximal linearization of a minimum phase model

The model (38) is decomposed into a non-controllable system and a controllable one as it was done for model (35). The dynamics of the center of gravity CG is not affected by the control input which means that CG has a similar ballistic motion as in Fig. 13. These non-controllable variables span the autonomous subspace $\mathcal{H}_\infty$ which reads as:

$$\mathcal{H}_\infty = \text{span}\{d\psi'_1, d\psi'_2\}$$

where

$$\begin{aligned} \psi'_1 &= \left(r_{p1}\dot{\theta} - \frac{m_3 + \bar{m}}{\bar{m}}v_3 \right)^2 + \left(\dot{r}_{p1} + r_{p3}\dot{\theta} + \frac{m_1 + \bar{m}}{\bar{m}}v_1 \right)^2, \\ \psi'_2 &= \left(r_{p1}\dot{\theta} - \frac{m_3 + \bar{m}}{\bar{m}}v_3 \right) \cos\theta \\ &\quad + \left(\dot{r}_{p1} + r_{p3}\dot{\theta} + \frac{m_1 + \bar{m}}{\bar{m}}v_1 \right) \sin\theta. \end{aligned}$$

There is now only one single independent function $\tilde{\phi}'_2$ which has relative degree 3 for the model (38). More precisely, compute $\mathcal{H}_3$:

$$\mathcal{H}_3 = \text{span}\{d\tilde{\phi}'_2, \omega\} \oplus \mathcal{H}_\infty$$

where

$$\tilde{\phi}'_2 = \theta + \frac{r_{p3}}{\sqrt{\frac{m_3(m_1+\bar{m})}{m_1(m_3+\bar{m})}} \sqrt{\frac{\bar{m}+m_1}{\bar{m}m_1}J + r_{p3}^2}} \arctan \frac{\sqrt{\frac{m_3(m_1+\bar{m})}{m_1(m_3+\bar{m})}} r_{p1}}{\sqrt{\frac{\bar{m}+m_1}{\bar{m}m_1}J + r_{p3}^2}}$$

and $\mathcal{H}_3$ is no more fully integrable. ω denotes a suitable non-integrable differential one-form.

Compute $\mathcal{H}_2$ as:

$$\mathcal{H}_2 = \text{span}\{dy'_1, dy'_2, d\tilde{\phi}'_1\} \oplus \mathcal{H}_\infty$$

where

$$y'_1 = \frac{\bar{m}}{m_3 + \bar{m}} r_{p1}\dot{\theta} - v_3,$$

$$y'_2 = \frac{\bar{m}}{m_1 + \bar{m}} (\dot{r}_{p1} + r_{p3}\dot{\theta}) + v_1,$$

$$\tilde{\phi}'_1 = J_2\dot{\theta} + \frac{\bar{m}m_1}{\bar{m} + m_1} \left(\frac{m_3(m_1 + \bar{m})}{m_1(m_3 + \bar{m})} r_{p1}^2\dot{\theta} + r_{p3}^2\dot{\theta} + r_{p3}\dot{r}_{p1} \right).$$

As in the previous sections, $\tilde{\phi}'_1$ is an angular momentum, related to $\tilde{\phi}'_2$ by

$$\dot{\tilde{\phi}'_2} = \frac{\frac{\bar{m}+m_1}{\bar{m}m_1}}{\frac{\bar{m}+m_1}{\bar{m}m_1}J_2 + \frac{m_3(m_1+\bar{m})}{m_1(m_3+\bar{m})}r_{p1}^2 + r_{p3}^2} \tilde{\phi}'_1.$$

Despite these dramatic changes in comparison to Section 4.3, it is now argued that the control scheme which has been computed for the fundamental special case in Section 4.2 is still valid, up to some approximation. This is done next.

4.4.3. Control design and simulation

Since $\tilde{\phi}'_1$ does not have relative degree 3 as $\tilde{\phi}'_2$, the results in Theorems 1 and 2 are not available anymore. However, the second time-derivative of $\tilde{\phi}'_1$ is argued to depend weakly on the control input u , so the coefficient of u is neglected.

More precisely, define again the output $y = \tilde{\phi}'_1 + k\tilde{\phi}'_2$, for some $k > 0$.

Instead of the coordinate (26)–(29), define a new coordinate as:

$$\xi_1 = \tilde{\phi}'_1 + k\tilde{\phi}'_2, \quad (39)$$

$$\xi_2 = \dot{\tilde{\phi}'_1} + k\dot{\tilde{\phi}'_2}, \quad (40)$$

$$\xi_3 = \ddot{\tilde{\phi}'_1} + k\ddot{\tilde{\phi}'_2} - \Delta u, \quad (41)$$

$$\zeta = \tilde{\phi}'_1 \quad (42)$$

where $\ddot{\tilde{\phi}'_1} = \Pi + \Delta u$ for some functions of the state variables Π and Δ .

Instead of solving Eq. (31), consider the following equation:

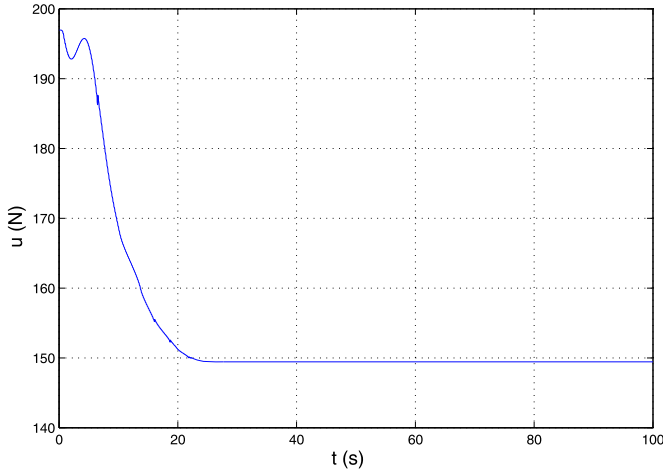


Fig. 14. The behavior of input u .

$$\dot{\xi}_3 + \lambda_2 \xi_3 + \lambda_1 y^{(1)} + \lambda_0 (y - y_e) = 0. \quad (43)$$

Note that Eq. (43) corresponds strictly to Eq. (31) only if $\Delta = 0$. In the following when Δ is small, then (43) is an approximation of (31).

In (43), one has $\dot{\xi}_3 = \alpha'(\xi, \zeta) + \beta'(\xi, \zeta)u$. Solving (43) in u , yields

$$u = -\frac{1}{\beta'(\xi, \zeta)} (\lambda_2 \xi_3 + \lambda_1 \xi_2^{(1)} + \lambda_0 (\xi_1 - y_e) + \alpha'(\xi, \zeta))$$

which is applied to the model (38). For the same reason, the explicit expression of u is not displayed here. $m_1 = 400$ kg, $m_3 = 500$ kg and the other parameters have the same values as previously. The simulation results of the dynamics of θ , r_{p1} and the motion of CG are similar to the previous cases. The behavior of input u is displayed in Fig. 14.

4.5. The complete airship model in the vertical plane

In the last step, the aerodynamic forces are added to the system, which means that now all the assumptions are dropped. The addition of the aerodynamic forces causes a huge impact on the angular momentum, since a pitch moment is included by the aerodynamic forces.

Combining (4), (14), (15), (16), and (17) the complete model in the vertical plane is derived, which also can be obtained from the full model (19), as follows

$$\begin{cases} \ddot{\theta} = \varrho_2 + \frac{M_a}{J + \bar{m}r_{p1}^2}, \\ \ddot{r}_{p1} = \sigma_2 + \frac{M_a r_{p3}}{J + \bar{m}r_{p1}^2}, \\ \dot{v}_1 = \kappa_1 + \frac{F_{a1}}{m_1}, \\ \dot{v}_3 = \kappa_3 + \frac{F_{a3}}{m_3 + \bar{m}} \end{cases}$$

where

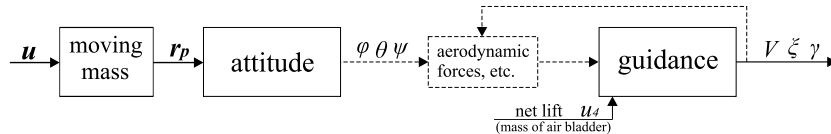


Fig. 15. The model structure of the buoyancy-driven airship.

$$F_{a1} = -X_a \cos \alpha + Z_a \sin \alpha,$$

$$F_{a3} = -X_a \sin \alpha - Z_a \cos \alpha.$$

There is no ballistic motion anymore, and the system becomes fully accessible:

$$\mathcal{H}_\infty = 0.$$

The aerodynamic forces render all DOF controllable. For instance, the special cases in Sections 4.3 and 4.4 are not fully controllable as no aerodynamic forces are considered, but the system in this section is fully accessible as all forces are considered. The airship's full model can be represented in Fig. 15.

The attitude of the airship is controlled by the position of the moving mass (r_p). Meanwhile, its position r_p is controlled by the internal forces u . The guidance of the airship is impacted by the net lift (controlled by u_4), aerodynamic forces and torques, and so on. The aerodynamic forces and torques introduce a new coupling which yields full accessibility, but they do not yield any new DOF. For the cases in Sections 4.3 and 4.4, due to the absence of the aerodynamic force, the center of gravity CG of the airship is subject to an uncontrollable ballistic motion, but in this section where the aerodynamic force is considered, the motion of CG is controllable, and is shown in the next simulations.

Compute $\mathcal{H}_2$ and $\mathcal{H}_3$ as:

$$\mathcal{H}_2 = \text{span}\{dy'_1, dy'_2, d\tilde{\phi}'_1, d\theta, dr_{p1}\},$$

$$\mathcal{H}_3 = \text{span}\{d\tilde{\phi}'_2, \omega_1, \omega_2, \omega_3\}.$$

Here, $d\tilde{\phi}'_1$ and $d\tilde{\phi}'_2$ are the same as those in Section 4.4. ω_1 , ω_2 , and ω_3 are suitable possibly non-integrable differential one-forms.

As in Section 4.4, an advanced nonlinear control u is derived which achieves internal stabilization. But due to its length, the explicit expression of u is not displayed here.

The simulation results of this nonlinear controller for the complete system are displayed in Figs. 16, 17. The initial values for θ , $\dot{\theta}$, r_{p1} , $\dot{r}_{p1}$, v_1 , and v_3 are 41.5° , 0, -2.5 m, 0, 10 m/s, and -0.69 m/s, respectively. The equilibrium values for these states are 30° , 0, -2.7 m, 0, 10.4 m/s, and -0.73 m/s, respectively. The angle of attack $\alpha = \theta - \xi = \arctan \frac{v_3}{v_1} = -4^\circ$. In this situation, the airship does not display a ballistic motion anymore, but flies along the desired angle which is denoted by ξ , here, $\xi = 34^\circ$ (see Fig. 19). Fig. 18 reflects the value of the control input u , which keeps in a reasonable domain. The main difference among the inputs of different cases is that the largest instant value is increasing, from 175 N of the first case to 220 N of the complete case. That is caused by the additions of forces to the system, such as inertial forces and aerodynamic forces. As depicted in the above, the input u takes the responsibility for the position of the moving mass, and the position of the moving mass is decided by the pitch angle. So, for the similar values of the pitch angles in different case, the final values of inputs are similar, around 150 N.

4.6. Simulation of a typical trajectory tracking flight

In this section, a typical trajectory is considered, as depicted in Fig. 5. It includes two cycles of ascending and descending segments. The goal is to show that the control laws developed in

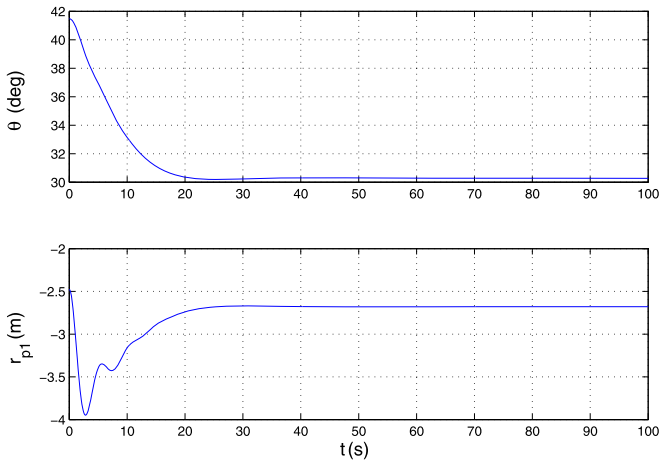


Fig. 16. Dynamics of θ and r_{p1} with initial error of the complete model.

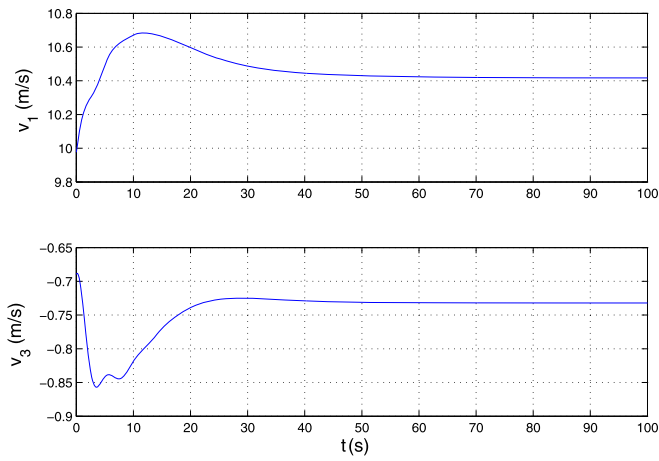


Fig. 17. Dynamics of v_1 and v_3 with initial error of the complete model.

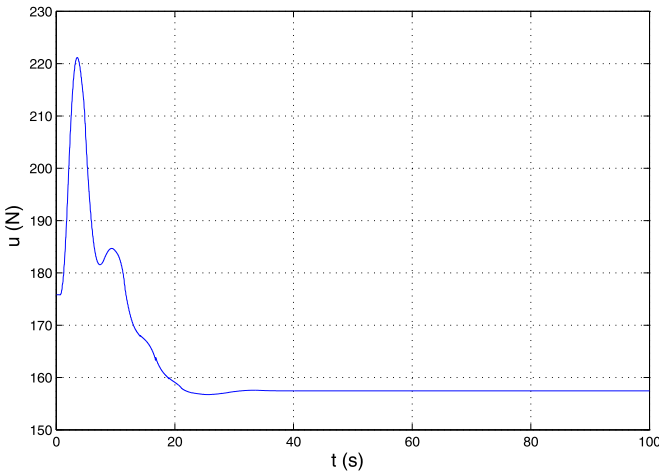


Fig. 18. The behavior of input u .

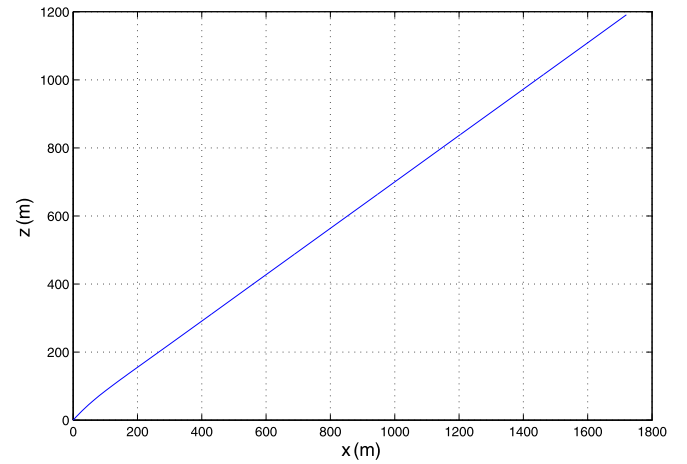


Fig. 19. The movement of O .

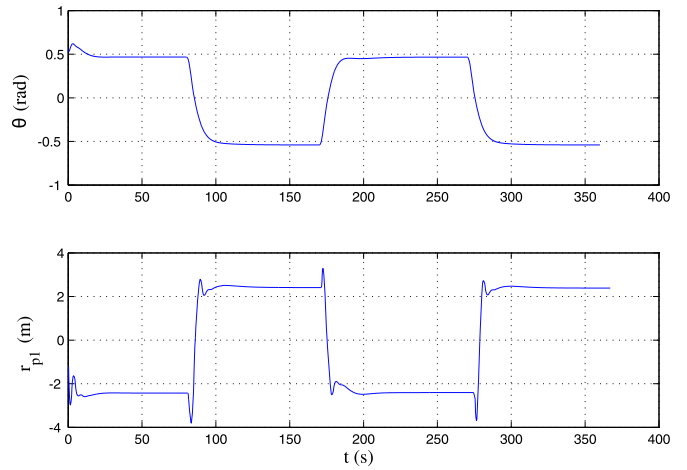


Fig. 20. Dynamics of θ and r_{p1} of a continuing flight.

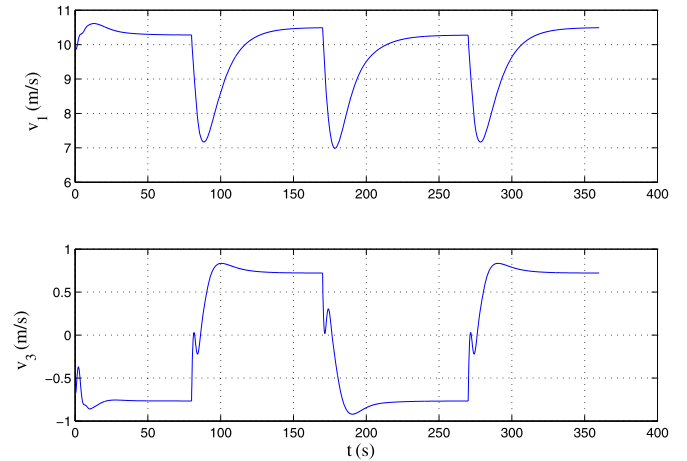


Fig. 21. Dynamics of v_1 and v_3 of a continuing flight.

Section 4.5 are able to cope with a switching structure of the system. To implement a sawtooth flight as Fig. 5, besides the input in Section 4.5 to control the movement of the moving mass, the input u_4 in Eq. (18) is involved to control the ascent and the descent.

The flight path is shown in Fig. 23, the variations of θ , r_{p1} , v_1 , v_3 are shown in Figs. 20 and 21. The input force is shown in Fig. 22.

The control laws for ascending and descending segments are similar, up to the desired command y_e . On the transition points, besides the switches of the commands of the states, the net buoyancy of the airship is switching from 340 N to -340 N, which represent a bang–bang control for u_4 . In this simulation, it is assumed that the release and the inflation of the air bladder are much faster than the motion of the airship, and this switch is considered to be instantaneous. It is shown by the simulation that

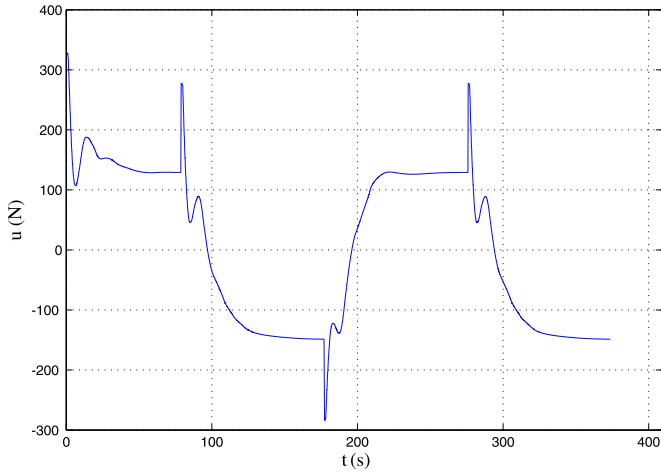
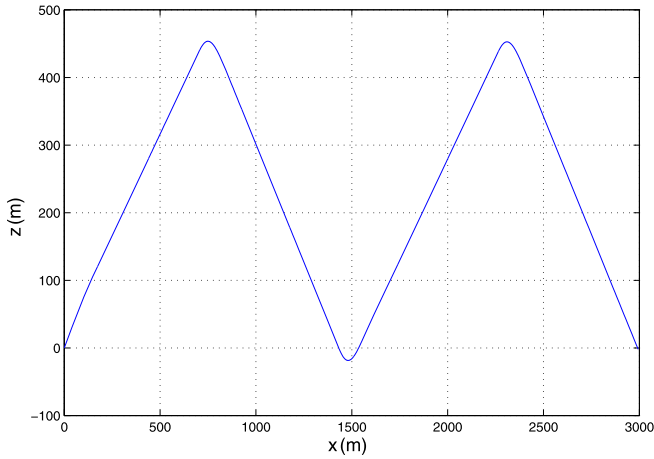
Fig. 22. The behavior of input u .

Fig. 23. Flight path of two cycles.

the behavior of the states and the control input is approximately symmetric. On the transition point, the variation of the input is acceptable as well.

5. Conclusion

The autonomous buoyancy-driven airship which uses an internal moving mass to control the attitude and an adjustable air bladder to control the altitude offers a novel mechanism for UAVs. Its mathematical model is nonlinear, and it is shown to be not fully-feedback linearizable. A new control methodology is developed based on maximal feedback linearization with stable internal dynamics. Thus, the approach and the result presented in this paper is instrumental for the analysis of similar mechanical systems, such as underwater glider and reentry vehicle.

Through the Newton–Euler approach, a 3D full model with 8-DOF is derived, which is more complicated than conventional autonomous airship due to the existence of the internal dynamics and special lift propeller. This paper views the internal moving mass and the rigid body of the vehicle as two independent parts, which renders the modeling clear and easy, and also displays the interconnection of the system. The analysis in Section 4.5 shows the role of the aerodynamic forces for the dynamics of this airship, which introduces a coupling between the attitude control and the guidance control and yields full accessibility of the system's DOF.

To deal with a complex nonlinear mechanical system, this paper offers a new approach which is to analyze the system with a maximal number of constraints, which are gradually relaxed. The max-

imally constrained model is assimilated to be the prismatic-joint pendulum shown in Section 4.2. Thanks to the angular momentum of the whole airship computed at the point O or CG which is denoted by ϕ_1 , $\dot{\phi}_1$ and ϕ_1' in this paper, explicit output functions which define a minimum phase system are derived. Maximal feedback linearization with internal stability is performed, and good control performance is obtained.

The modeling approach, dynamic analysis and control method displayed in this paper make sense for similar complex systems, and is also instrumental in dealing with other motions of this airship, for instance, the dynamics in the horizontal plane, which is open for future work. Other research perspectives include the consideration of wind.

The final claim is that the control designed in this paper is a first step for a general realistic control scheme for this airship and deserves to be combined for instance with some robust control methods valid for the linearized system.

Appendix A

The explicit expression of u in Eq. (32) is as follows:

$$u = -\frac{1}{\beta}(\lambda_2 \xi_3^{(2)} + \lambda_1 \xi_2^{(1)} + \lambda_0(\xi_1 - y_e) + \alpha)$$

where $\alpha = \alpha_1 + \alpha_2$ and $\beta = \beta_1 + \beta_2$, and they are computed as follows:

$$\phi_1^{(3)} = \alpha_1 + \beta_1 u,$$

$$\alpha_1 = -g(-\cos\theta g \sin\theta m J - 2m\dot{r}_{p1} \sin\theta \dot{\theta} J - mr_{p3} \sin\theta \dot{\theta}^2 J) / (J + mr_{p1}^2),$$

$$\beta_1 = (\cos\theta J + \cos\theta mr_{p1}^2 + mr_{p1} \sin\theta r_{p3}) / (J + mr_{p1}^2),$$

$$\phi_2^{(3)} = \alpha_2 + \beta_2 u,$$

$$\begin{aligned} \alpha_2 = & -(2\dot{\theta}^3 \bar{m}^4 r_{p1}^8 + 2\bar{m}^3 \dot{r}_{p1}^2 r_{p3}^4 \dot{\theta} J + 4\bar{m}^2 \dot{r}_{p1}^2 r_{p3}^2 \dot{\theta} J^2 \\ & + \bar{m} g r_{p3} \cos(\theta) \dot{\theta} J^3 + \bar{m}^3 \dot{r}_{p1} g \cos(\theta) r_{p3}^4 J \\ & + \bar{m}^3 g r_{p3}^5 \cos(\theta) \dot{\theta} J + 2\bar{m}^2 g r_{p3}^3 \cos(\theta) \dot{\theta} J^2 \\ & + \bar{m} \dot{r}_{p1} g J^3 \cos(\theta) - 3r_{p1}^7 g \sin(\theta) \dot{\theta} \bar{m}^4 \\ & + 3r_{p1}^6 \bar{m}^4 \dot{\theta} g r_{p3} \cos(\theta) - 3r_{p1}^6 \bar{m}^4 g \cos(\theta) \dot{r}_{p1} \\ & + 6r_{p1}^6 \bar{m}^4 \dot{\theta}^2 r_{p3} \dot{r}_{p1} - 6r_{p1}^5 \bar{m}^4 g \sin(\theta) \dot{\theta} r_{p3}^2 \\ & - 6r_{p1}^5 \bar{m}^4 r_{p3} g \sin(\theta) \dot{r}_{p1} - 9r_{p1}^5 \bar{m}^3 g \sin(\theta) \dot{\theta} J \\ & + 12r_{p1}^4 \bar{m}^4 \dot{r}_{p1} r_{p3}^3 \dot{\theta}^2 + 2\bar{m}^2 \dot{r}_{p1} g J^2 \cos(\theta) r_{p3}^2 \\ & + 6r_{p1}^4 \bar{m}^4 \dot{\theta} r_{p3}^3 g \cos(\theta) + 7r_{p1}^4 \bar{m}^3 g r_{p3} \cos(\theta) \dot{\theta} J \\ & - 10r_{p1}^4 \bar{m}^3 \dot{\theta}^2 \dot{\theta} J + 12r_{p1}^4 \bar{m}^3 \dot{\theta}^3 r_{p3}^2 J - 5r_{p1}^4 \bar{m}^3 \dot{r}_{p1} g J \cos(\theta) \\ & + 12r_{p1}^4 \bar{m}^3 \dot{\theta} r_{p3} \dot{\theta}^2 J - 3r_{p1}^3 \bar{m}^4 g \sin(\theta) \dot{\theta} r_{p3}^4 \\ & - 6r_{p1}^3 \bar{m}^4 r_{p3}^3 g \sin(\theta) \dot{\theta} - 12r_{p1}^3 \bar{m}^3 g \sin(\theta) \dot{\theta} r_{p3}^2 J \\ & - 12r_{p1}^3 \bar{m}^3 r_{p3} g \sin(\theta) \dot{\theta} J - 9r_{p1}^3 \bar{m}^2 g J^2 \sin(\theta) \dot{\theta} \\ & + 3r_{p1}^2 \bar{m}^4 \dot{\theta} r_{p3}^5 g \cos(\theta) + 6r_{p1}^2 \bar{m}^4 \dot{r}_{p1} r_{p3}^5 \dot{\theta}^2 \\ & + 3r_{p1}^2 \bar{m}^4 \dot{r}_{p1} g \cos(\theta) r_{p3}^4 - 4r_{p1}^2 \bar{m}^3 \dot{r}_{p1}^3 r_{p3} J \\ & + 12r_{p1}^2 \bar{m}^3 \dot{r}_{p1} r_{p3}^3 \dot{\theta}^2 J + 4r_{p1}^2 \bar{m}^3 \dot{r}_{p1}^2 r_{p3}^2 \dot{\theta} J \\ & + 2r_{p1}^2 \bar{m}^3 \dot{r}_{p1} g \cos(\theta) r_{p3}^2 J + 6r_{p1}^2 \bar{m}^3 \dot{\theta}^3 r_{p3}^4 J \\ & + 8r_{p1}^2 \bar{m}^3 g r_{p3}^3 \cos(\theta) \dot{\theta} J - 2r_{p1}^2 \bar{m}^2 \dot{r}_{p1}^2 \dot{\theta} J^2 \\ & + 6r_{p1}^2 \bar{m}^2 \dot{\theta}^3 r_{p3}^2 J^2 - r_{p1}^2 \bar{m}^2 \dot{r}_{p1} g J^2 \cos(\theta) \end{aligned}$$

$$\begin{aligned}
& + 5r_{p1}^2 \bar{m}^2 g r_{p3} \cos(\theta) \dot{\theta} J^2 + 6r_{p1}^2 \bar{m}^2 \dot{r}_{p1} r_{p3} \dot{\theta}^2 J^2 \\
& - 3r_{p1} \bar{m}^3 g \sin(\theta) \dot{\theta} r_{p3}^4 J - 6r_{p1} \bar{m}^2 r_{p3} g \sin(\theta) \dot{r}_{p1} J^2 \\
& - 3r_{p1} \bar{m} g J^3 \sin(\theta) \dot{\theta} - 6r_{p1} \bar{m}^3 r_{p3}^3 g \sin(\theta) \dot{r}_{p1} J \\
& + 2\bar{m}^3 J \dot{r}_{p1}^3 r_{p3}^3 + 2\bar{m}^2 \dot{r}_{p1}^3 r_{p3} J^2 + 2\bar{m} \dot{r}_{p1}^2 \dot{\theta} J^3 \\
& + 6r_{p1}^2 \bar{m}^4 \dot{r}_{p1}^2 r_{p3}^4 \dot{\theta} + 6r_{p1}^6 \bar{m}^4 \dot{\theta}^3 r_{p3}^2 - 6r_{p1}^6 \bar{m}^4 \dot{r}_{p1}^2 \dot{\theta} \\
& + 6r_{p1}^6 \dot{\theta}^3 J \bar{m}^3 + 6r_{p1}^4 \bar{m}^4 \dot{\theta}^3 r_{p3}^4 - 6r_{p1}^4 \bar{m}^4 \dot{r}_{p1}^3 r_{p3} \\
& + 6r_{p1}^4 \dot{\theta}^3 J^2 \bar{m}^2 + 2r_{p1}^2 \bar{m}^4 \dot{r}_{p1}^3 r_{p3}^3 + 2r_{p1}^2 \bar{m}^4 \dot{\theta}^3 r_{p3}^6 \\
& + 2r_{p1}^2 \dot{\theta}^3 J^3 \bar{m} - 6r_{p1} \bar{m}^2 g J^2 \sin(\theta) \dot{\theta} r_{p3}^2 / (J^4 + 4J^3 \bar{m} r_{p1}^2 \\
& + 6J^2 \bar{m}^2 r_{p1}^4 + 9J \bar{m}^3 r_{p1}^4 r_{p3}^2 + 3J^2 \bar{m}^2 r_{p3}^4 + 6J \bar{m}^3 r_{p3}^4 r_{p1}^2 \\
& + \bar{m}^4 r_{p1}^8 + 3\bar{m}^4 r_{p1}^6 r_{p3}^2 + 3\bar{m}^4 r_{p1}^4 r_{p3}^4 + \bar{m}^3 r_{p3}^6 J \\
& + \bar{m}^4 r_{p3}^6 r_{p1}^2 + 9J^2 \bar{m}^2 r_{p3}^2 r_{p1}^2 + 4J \bar{m}^3 r_{p1}^6 + 3J^3 \bar{m} r_{p3}^2), \\
\beta_2 = & (2r_{p1} \dot{\theta} J^3 + 6r_{p1} \bar{m} \dot{\theta} r_{p3}^2 J^2 + 2r_{p1}^7 \dot{\theta} \bar{m}^3 + 2r_{p1} \bar{m} \dot{r}_{p1} r_{p3} J^2 \\
& + 6r_{p1} \bar{m}^2 \dot{\theta} r_{p3}^4 J + 4r_{p1} \bar{m}^2 \dot{r}_{p1} r_{p3}^3 J + 2r_{p1} \bar{m}^3 \dot{\theta} r_{p3}^6 \\
& + 2r_{p1} \bar{m}^3 \dot{r}_{p1} r_{p3}^5 + 6r_{p1}^3 \dot{\theta} J^2 \bar{m} + 6r_{p1}^5 \bar{m}^3 \dot{\theta} r_{p3}^2 \\
& + 2r_{p1}^5 \bar{m}^3 \dot{r}_{p1} r_{p3} + 6r_{p1}^5 \dot{\theta} J \bar{m}^2 + 6r_{p1}^3 \bar{m}^3 \dot{\theta} r_{p3}^4 + 4r_{p1}^3 \bar{m}^3 \dot{r}_{p1} r_{p3}^3 \\
& + 4r_{p1}^3 \bar{m}^2 \dot{r}_{p1} r_{p3} J + 12r_{p1}^3 \bar{m}^2 \dot{\theta} r_{p3}^2 J) / (J^4 + 4J^3 \bar{m} r_{p1}^2 \\
& + 6J^2 \bar{m}^2 r_{p1}^4 + 9J \bar{m}^3 r_{p1}^4 r_{p3}^2 + 3J^2 \bar{m}^2 r_{p3}^4 + 6J \bar{m}^3 r_{p3}^4 r_{p1}^2 \\
& + \bar{m}^4 r_{p1}^8 + 3\bar{m}^4 r_{p1}^6 r_{p3}^2 + 3\bar{m}^4 r_{p1}^4 r_{p3}^4 + \bar{m}^3 r_{p3}^6 J \\
& + \bar{m}^4 r_{p3}^6 r_{p1}^2 + 9J^2 \bar{m}^2 r_{p3}^2 r_{p1}^2 + 4J \bar{m}^3 r_{p1}^6 + 3J^3 \bar{m} r_{p3}^2).
\end{aligned}$$

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